OJT(On Job Training)-2

Honeypot Setup Project

College Name:-Government Polytechnic Udupi

Project Name:-Honeypot Setup

Project Team Members:-Koushik

Information:-

Honeypot Setup – Introduction

A honeypot is a cybersecurity tool designed to detect, divert, and analyze unauthorized access attempts. It acts as a decoy system that attracts attackers so security professionals can monitor their behavior and improve network defenses.

Honeypot Setup – Objective

The objective of this project is to deploy a honeypot system that can capture malicious activities such as unauthorized login attempts, malware uploads, and scanning attacks. This helps in understanding attacker techniques and improving overall cybersecurity.

Honeypot Setup – Tools Used

The honeypot environment is created using tools such as Kali Linux, Cowrie Honeypot, Python, and Virtual Machines. These tools help simulate vulnerable services and record attacker activities for analysis.

Honeypot Setup – Working Principle

A honeypot works by imitating real services like SSH, FTP, or HTTP. When an attacker attempts to access these services, the honeypot records their commands, IP address, login attempts, and activities without allowing them to damage the real system.

Key Components of Honeypot setup

Honeypot System – A decoy system that imitates real servers or services to attract attackers.

Monitoring System – Tracks and observes attacker activities in the honeypot.

Logging Mechanism – Records login attempts, commands, IP addresses, and other actions.

Analysis Tools – Helps security experts study attacker behavior and improve security.
Data Collection – Captures attack data for analysis and research.

Alert System – Sends notifications when suspicious activity is detected.

Purpose Of Honeypot Setup

A

Detect Cyber Attacks – To identify unauthorized access and hacking attempts in a network.

B

Analyze Attacker Behavior – To study the methods and techniques used by attackers.

C

Improve Network Security – To collect attack data and strengthen system defenses.

Honeypot Setup Code

```python
# Simple Login Honeypot (TCP) - Logs connection + username/password attempts
# Run: python honeypot.py
# Connect test:  telnet <your-ip> 2222   OR   nc <your-ip> 2222

import socket
import datetime

HOST = "0.0.0.0"
PORT = 2222
LOG_FILE = "honeypot_log.txt"

def log(msg: str):
    timestamp = datetime.datetime.now().strftime("%Y-%m-%d %H:%M:%S")
    line = f"[{timestamp}] {msg}"
    print(line)
    with open(LOG_FILE, "a", encoding="utf-8") as f:
        f.write(line + "\n")

def handle_client(conn: socket.socket, addr):
    ip, port = addr
    log(f"Connection from {ip}:{port}")

    try:
        conn.sendall(b"Welcome to Secure Shell Server v2.0\r\n")
        conn.sendall(b"Username: ")
        username = conn.recv(1024).strip().decode(errors="ignore")

        conn.sendall(b"Password: ")
        password = conn.recv(1024).strip().decode(errors="ignore")

        log(f"Login attempt from {ip}:{port} | username='{username}' password='{password}'")

        # Always fail (this is a decoy)
        conn.sendall(b"Access denied.\r\n")
    except Exception as e:
        log(f"Error with {ip}:{port} -> {e}")
    finally:
        conn.close()
        log(f"Disconnected {ip}:{port}")

def main():
    log(f"Starting honeypot on {HOST}:{PORT}")
    with socket.socket(socket.AF_INET, socket.SOCK_STREAM) as s:
        s.setsockopt(socket.SOL_SOCKET, socket.SO_REUSEADDR, 1)
        s.bind((HOST, PORT))
        s.listen(50)

        while True:
            conn, addr = s.accept()
            handle_client(conn, addr)

if __name__ == "__main__":
    main()
```

Purpose of the Honeypot Code

Capture Unauthorized Login Attempts
The code creates a fake login server that records usernames and passwords entered by attackers.

Monitor Attacker Activity
It logs the attacker’ s IP address, connection time, and login attempts for security analysis.

Cybersecurity Learning and Research
This honeypot helps students and security researchers understand how attackers try to access systems and improves network security strategies.

Conclusion (Honeypot Setup)

The honeypot setup helps in detecting and monitoring unauthorized access attempts in a network. It acts as a decoy system that attracts attackers and records their activities such as login attempts and commands. By analyzing this information, security professionals can understand attack patterns and improve the overall security of the system. Honeypots are useful tools for cybersecurity research, learning, and strengthening network defenses.

Thanks